

Modulation and Coding Schemes [MCS Index]

MCS Table (HT/VHT/HE) - MCSINDEX.NET

MCS Index				Spatial Stream	Modulation	Coding	OFDM (Prior 11ax)								OFDM & OFDMA (Starting with 11ax)																														
							20MHz		40MHz		80MHz		160MHz		26-tone RU				52-tone RU				106-tone RU				242-tone RU / 20MHz				484-tone RU / 40MHz				996-tone RU / 80MHz				2x996-tone RU / 160MHz						
HT	VHT	HE	EHT				0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI							
0	0	0	0	1	BPSK	1/2	6.5	7.2	13.5	15	29.3	32.5	58.5	65	0.8	0.8	0.9	1.5	1.7	1.8	3.2	3.5	3.8	7.3	8.1	8.6	14.6	16.3	17.2	30.6	34.0	36.0	61.3	68.1	72.1										
1	1	1	1	1	QPSK	1/2	13	14.4	27	30	58.5	65	117	130	1.5	1.7	1.8	3.0	3.3	3.5	6.4	7.1	7.5	14.6	16.3	17.2	29.3	32.5	34.4	61.3	68.1	72.1	122.5	136.1	144.1										
2	2	2	2	1	QPSK	3/4	19.5	21.7	40.5	45	87.8	97.5	175.5	195	2.3	2.5	2.6	4.5	5.0	5.3	9.6	10.6	11.3	21.9	24.4	25.8	43.9	48.8	51.6	91.9	102.1	108.1	183.8	204.2	216.2										
3	3	3	3	1	16-QAM	1/2	26	28.9	54	60	117	130	234	260	3.0	3.3	3.5	6.0	6.7	7.1	12.8	14.2	15.0	29.3	32.5	34.4	58.5	65.0	68.8	122.5	136.1	144.1	245.0	272.2	288.2										
4	4	4	4	1	16-QAM	3/4	39	43.3	81	90	175.5	195	351	390	4.5	5.0	5.3	9.0	10.0	10.6	19.1	21.3	22.5	43.9	48.8	51.6	87.8	97.5	103.2	183.8	204.2	216.2	367.5	408.3	432.4										
5	5	5	5	1	64-QAM	2/3	52	57.8	108	120	234	260	468	520	6.0	6.7	7.1	12.0	13.3	14.1	25.5	28.3	30.0	58.5	65.0	68.8	117.0	130.0	137.6	245.0	272.2	288.2	490.0	544.4	576.5										
6	6	6	6	1	64-QAM	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585	6.8	7.5	7.9	13.5	15.0	15.9	28.7	31.9	33.8	65.8	73.1	77.4	131.6	146.3	154.9	275.6	306.3	324.3	551.3	612.5	648.5										
7	7	7	7	1	64-QAM	5/6	65	72.2	135	150	292.5	325	585	650	7.5	8.3	8.8	15.0	16.7	17.6	31.9	35.4	37.5	73.1	81.3	86.0	146.3	162.5	172.1	306.3	340.3	360.3	612.5	680.6	720.6										
8	8	8	8	1	256-QAM	3/4	78	86.7	162	180	351	390	702	780	9.0	10.0	10.6	18.0	20.0	21.2	38.3	42.5	45.0	87.8	97.5	103.2	175.5	195.0	206.5	367.5	408.3	432.4	735.0	816.7	864.7										
9	9	9	9	1	256-QAM	5/6	N/A	N/A	180	200	390	433.3	780	866.7	10.0	11.1	11.8	20.0	22.2	23.5	42.5	47.2	50.0	97.5	108.3	114.7	195.0	216.7	229.4	408.3	453.7	480.4	816.7	907.4	960.8										
10	10	1	1	1024-QAM	3/4										11.3	12.5	13.2	22.5	25.0	26.5	47.8	53.1	56.3	109.7	121.9	129.0	219.4	243.8	258.1	459.4	510.4	540.4	918.8	1020.8	1080.9										
11	11	1	1	1024-QAM	5/6										12.5	13.9	14.7	25.0	27.8	29.4	53.1	59.0	62.5	121.9	135.4	143.4	243.8	270.8	286.8	510.4	567.1	600.5	1020.8	1134.3	1201.0										
12	12	1	1	4096-QAM	3/4										13.5	15.0	15.9	27.0	30.0	31.8	57.4	63.8	67.5	131.6	146.3	154.9	263.3	292.5	309.7	551.3	612.5	648.5	1102.5	1225.0	1297.1										
13	13	1	1	4096-QAM	5/6										15.0	16.7	17.6	30.0	33.3	35.3	63.8	70.8	75.0	146.3	162.5	172.1	292.5	325.0	344.1	612.5	680.6	720.6	1225.0	1361.1	1441.2										
14	14	0	0	0	2	BPSK	1/2	13	14.4	27	30	58.5	65	117	130	1.5	1.7	1.8	3.0	3.3	3.5	6.4	7.1	7.5	14.6	16.3	17.2	29.3	32.5	34.4	61.3	68.1	72.1	122.5	136.1	144.1									
15	15	1	1	1	2	QPSK	1/2	26	28.9	54	60	117	130	234	260	3.0	3.3	3.5	6.0	6.7	7.1	12.8	14.2	15.0	29.3	32.5	34.4	58.5	65.0	68.8	122.5	136.1	144.1	245.0	272.2	288.2									
16	16	2	2	2	2	QPSK	3/4	39	43.3	81	90	175.5	195	351	390	4.5	5.0	5.3	9.0	10.0	10.6	19.1	21.3	22.5	43.9	48.8	51.6	87.8	97.5	103.2	183.8	204.2	216.2	367.5	408.3	432.4									
17	17	3	3	3	3	16-QAM	1/2	52	57.8	108	120	234	260	468	520	6.0	6.7	7.1	12.0	13.3	14.1	25.5	28.3	30.0	58.5	65.0	68.8	117.0	130.0	137.6	245.0	272.2	288.2	490.0	544.4	576.5									
18	18	4	4	4	2	16-QAM	3/4	78	86.7	162	180	351	390	702	780	9.0	10.0	10.6	18.0	20.0	21.2	38.3	42.5	45.0	87.8	97.5	103.2	175.5	195.0	206.5	367.5	408.3	432.4	735.0	816.7	864.7									
19	19	5	5	5	2	64-QAM	2/3	104	115.6	216	240	468	520	936	1040	12.0	13.3	14.1	24.0	26.7	28.2	51.0	56.7	60.0	117.0	130.0	137.6	234.0	260.0	275.3	490.0	544.4	576.5	980.0	1088.9	1152.9									
20	20	6	6	6	2	64-QAM	3/4	117	130	243	270	526.5	585	1053	1170	13.5	15.0	15.9	27.0	30.0	31.8	57.4	63.8	67.5	131.6	146.3	154.9	263.3	292.5	309.7	551.3	612.5	648.5	1102.5	1225.0	1297.1									
21	21	7	7	7	2	64-QAM	5/6	130	144.4	270	300	585	650	1170	1300	15.0	16.7	17.6	30.0	33.3	35.3	63.8	70.8	75.0	146.3	162.5	172.1	292.5	325.0	344.1	612.5	680.6	720.6	1225.0	1361.1	1441.2									
22	22	8	8	8	2	256-QAM	3/4	156	173.3	324	360	702	780	1404	1560	18.0	20.0	21.2	36.0	40.0	42.4	76.5	85.0	90.0	175.5	195.0	206.5	351.0	390.0	412.9	735.0	816.7	864.7	1470.0	1633.3	1729.4									
23	23	9	9	9	2	256-QAM	5/6	N/A	N/A	360	400	780	866.7	1560	1733.3	20.0	22.2	23.5	40.0	44.4	47.1	85.0	94.4	100.0	195.0	216.7	229.4	390.0	433.3	458.8	816.7	907.4	960.8	1633.3	1814.8	1921.6									
24	24	10	10	2	1024-QAM	3/4									22.5	25.0	26.5	45.0	50.0	52.9	95.6	106.3	112.5	219.4	243.8	258.1	438.8	487.5	516.2	918.8	1020.8	1080.9	1837.5	2041.7	2161.8										
25	25	11	11	2	1024-QAM	5/6									25.0	27.8	29.4	50.0	55.6	58.8	106.3	118.1	125.0	243.8	270.8	286.8	487.5	541.7	573.5	1020.8	1134.3	1201.0	2041.7	2268.5	2402.0										
26	26	12	12	2	4096-QAM	3/4									27.0	30.0	31.8	54.0	60.0	63.5	114.8	127.5	135.0	263.3	292.5	309.7	526.5	585.0	619.4	1102.5	1225.0	1297.1	2205.0	2450.0	2594.1										
27	27	13	13	2	4096-QAM	5/6									30.0	33.3	35.3	60.0	66.7	70.6	127.5	141.7	150.0	292.5	325.0	344.1	585.0	650.0	688.2	1225.0	1361.1	1441.2	2450.0	2722.2	2882.4										
28	28	0	0	0	3	BPSK	1/2	19.5	21.7	40.5	45	87.8	97.5	175.5	195	2.3	2.5	2.6	4.5	5.0	5.3	9.6	10.6	11.3	21.9	24.4	25.8	43.9	48.8	51.6	91.9	102.1	108.1	183.8	204.2	216.2									
29	29	1	1	1	3	QPSK	1/2	39	43.3	81	90	175.5	195	351	390	4.5	5.0	5.3	9.0	10.0	10.6	19.1	21.3	22.5	43.9	48.8	51.6	87.8	97.5	103.2	183.8	204.2	216.2	367.5	408.3	432.4									
30	30	2	2	2	3	QPSK	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585	6.8	7.5	7.9	13.5	15.0	15.9	28.7	31.9	33.8	65.8	73.1	77.4	131.6	146.3	154.9	275.6	306.3	324.3	551.3	612.5	648.5									
31	31	3	3	3	3	16-QAM	1/2	78	86.7	162	180	351	390	702	780	9.0	10.0	10.6	18.0	20.0	21.2	38.3	42.5	45.0	87.8	97.5	103.2	175.5	195.0	206.5	367.5	408.3	432.4	735.0	816.7	864.7									
32	32	4	4	4	3	16-QAM	3/4	117	130	243	270	526.5	585	1053	1170	13.5	15.0	15.9	27.0	30.0	31.8	57.4	63.8	67.5	131.6	146.3	154.9	263.3	292.5	309.7	551.3	612.5	648.5	1102.5	1225.0	1297.1									
33	33	5	5	5	3	64-QAM	2/3	156	173.3	324	360	702	780	1404	1560	18.0	20.0	21.2	36.0	40.0	42.4	76.5	85.0	90.0	175.5	195.0	206.5	351.0	390.0	412.9	735.0	816.7	864.7	1470.0	1633.3	1729.4									
34	34	6	6	6	3	64-QAM	3/4	175.5	195	364.5	405	N/A	N/A	15																															